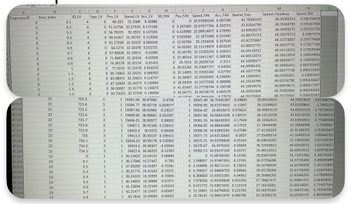
vicolungo.csv

→ Original Source
Open Acc Dataset
→ maintained by JRC
European Commissions
Joint Research

It's purpose is specifically to provide real car following experiments involving vehicles equipped with ACC (Adaptive Cruise Control).



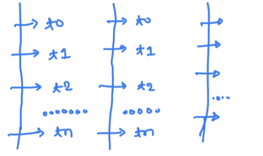
This file contains :-

- 158,018 → samples
- 61 trajectories
- 10Hz sampling

ie 10 measurements per seconds

That means every $\Delta t = 0.1$sec we have another observation of the vehicle.

traj 0 traj 1 traj 61



Id_LV → LV = Lead Vehicle

These are the vehicle identities associated with the OpenACC vicolungo Dataset.

ID	vehicle
0	Ford S-max
1	Kia Niro
2	Mini Cooper
....	
6	VW Golf E

e.g. ID_LV = 0 ; Ford S-Max

Type_LV = 1 ; that Ford S-Max is an "automated vehicle".

Type_LV = 0 ; the lead vehicle is "human_driven".

- Spatial_Gap ⇒ It represents bumper-to-bumper distance between two vehicles.

$g = P_{LV} - P_{FAV} - \frac{L_{FAV}}{2} - \frac{L_{LV}}{2}$ P_{LV} = position of lead vehicle. L_W = location of lead vehicle.

- Spatial Headway ⇒ It represents the distance between centers of the two vehicles.

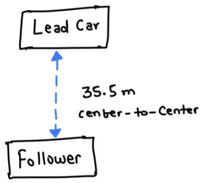
$h_{it} = P^f_{it} - P^f_{it}$

- Speed_Diff ⇒ $\Delta V = V_{LV} - V_{FAV}$ m/s

eg ① $V_{LV} = 25$ $V_{FAV} = 30$ $\Delta V = +5$m/s The lead is pulling away.

② $V_{LV} = 20$ $V_{FAV} = 25$ $\Delta V = -5$m/s The follower is faster.

One Simple Trajectory Example :-



Suppose, at one instant,

- lead position (P_{LV}) = 50m
- follower posn (P_{FAV}) = 10m
- both vehicle lengths = 4.5m
- lead speed = 19 m/s
- follower speed = 20 m/s

① $g = P_{LV} - P_{FAV} - \frac{L_{FAV}}{2} - \frac{L_{LV}}{2}$

$\Rightarrow 50 - 10 - \frac{4.5}{2} - \frac{4.5}{2}$

$g \Rightarrow 35.5$m

So, the cars actual physical gap is $g = 35.5$m

② Calculate relative velo ; $\Delta V = V_{LV} - V_{FAV} \Rightarrow 19 - 20$

$\Rightarrow -1$ m/s

(follower is faster → car's are closing)

③ Now CBF defines what safe means !

$h = g - (T_{min} V_{FAV} + d_{min})$

↓ =2s ↓ =15m

$h = -19.5$; so, according to CBF safety condition ⇒ $h \geq 0$

★ We are currently inside the Unsafe barrier -region

CBF now calculates the maximum allowed accele-ration $\overset{\circ}{h}$ →

$$\overset{\circ}{h} = \Delta V - T_{min}\, u$$

CBF condn ⇒ $\overset{\circ}{h} + kh \geq 0$

$$\therefore u \leq \frac{kh + \Delta V}{T_{min}}$$

using $k = 1$, $u_{CBF} = \frac{0.1(-19.5) + (-1)}{2}$

$\Rightarrow \frac{-2.95}{2} \Rightarrow u_{CBF} = -1.475\ m/s^2$

u CBF is basically saying. Given the current gap, speed & closing speed, don't accelerate more than this. You need to break atleast enough to satisfy safety constraint. u,

But we also have a normal ACC Controller

The vehicle doesn't normally want to break just because CBF exists.

It first calculates a nominal ACC Command.

Our prototype 2 nominal controller was based on

$S_{desired} = T_{des} V + D_{des}$

↓ = 1.5s ↓ =5m

Conceptually :

ACC " What Acceleration would I normally like ?"

CBF " What Acceleration is allowed for Safety ?"

Combine them ? The key controller from the paper is :- $u = \min(u_{nom}, u_{CBF})$

e.g. $u_{nom} = -0.7$ & $u_{CBF} = -1.475$

Then $u_{min} = -1.475\ m/s^2$

So : Normal ACC wants : $-0.7\ m/s^2$

↓

CBF says maximum : $-1.475\ m/s^2$

↓

final command : $-1.475\ m/s^2$

↓

BRAKE

Then the simulation moves forward —

t = 0

↓

read lead / follower stats

↓

calculate gap

↓

calculate ΔV

↓

calculate h

↓

calculate u_{CBF}

↓

calculate u_{nom}

↓

choose final u

↓

apply acceleration

↓

t = 0.05sec

↓

REPEAT